33. **a.** No **b.** No **c.** No

34. **a.** No **b.** Yes **c.** No

35. **a.** Yes, 180° **b.** Yes **c.** No

36. **a.** Yes, 60°, 120°, and 180° **b.** Yes **c.** No

37. **a.** Yes, 120° **b.** Yes **c.** No

38. **a.** No **b.** No **c.** Yes **d.** (1, 0) and (0, 1)

39. **a.** Yes, 90° and 180° **b.** Yes **c.** No **d.** (1, 1) and (-1,1)

40. **a.** Yes, 120° **b.** No **c.** No **d.** (1,0) and $(1, \sqrt{3})$

41. **a.** Yes, 120° **b.** Yes **c.** No **d.** (0, 1) and $(\sqrt{3}, 1)$

42. Let us call the four diagonals of the cube through its center d_1, d_2, d_3, and d_4. By rotating the cube, any diagonal can be moved to fall on the line segment formerly occupied by any of the diagonals (including itself) in two ways. For example, if d_1 goes from point P to point Q and d_2 from point R to point S, then d_1 can be moved into the segment from R to S with the vertex formerly at P falling on either point R or point S. Thus diagonal d_1 can be moved onto a diagonal (including itself) in $4 \cdot 2 = 8$ ways. Once diagonal d_1 is in position, we can keep the ends of d_1 fixed and rotate the cube through a total of three positions, giving a total of $3 \cdot 8 = 24$ possible rotations of the cube. But the set $\{d_1, d_2, d_3, d_4\}$ admits only $4! = 24$ permutations. Thus, identifying each rotation with one of these permutations of the four diagonals, we see that the group of rotations must be isomorphic to the full symmetric group S_4 on four letters.

13. Homomorphisms

1. It is a homomorphism, because $\phi(m + n) = m + n = \phi(m) + \phi(n)$.

2. It is not a homomorphism, because $\phi(2.6 + 1.6) = \phi(4.2) = 4$ but $\phi(2.6) + \phi(1.6) = 2 + 1 = 3$.

3. It is a homomorphism, because $\phi(xy) = |xy| = |x||y| = \phi(x)\phi(y)$ for $x, y \in \mathbb{R}^*$.

4. It is a homomorphism. Let $m, n \in \mathbb{Z}_6$. In $\mathbb{Z}$, let $m + n = 6q + r$ by the division algorithm in $\mathbb{Z}$. Then $\phi(m +_6 n)$ is the remainder of r modulo 2. Because 2 divides 6, the remainder of $m + n$ in $\mathbb{Z}$ modulo 2 is also the remainder of r modulo 2. Now the map $\gamma : \mathbb{Z} \to \mathbb{Z}_2$ of Example 13.10 is a homomorphism, and we have just shown that $\phi(m +_6 n) = \gamma(m + n)$ for $m, n \in \mathbb{Z}_6$. Thus we have $\phi(m +_6 n) = \gamma(m + n) = \gamma(m) +_2 \gamma(n) = \phi(m) + \phi(n)$.

5. It is not a homomorphism, because $\phi(5 +_9 7) = \phi(3) = 1$ but $\phi(5) +_2 \phi(7) = 1 +_2 1 = 0$.

6. It is a homomorphism, because $\phi(x + y) = 2^{x+y} = 2^x 2^y = \phi(x)\phi(y)$ for $x, y \in \mathbb{R}^*$.

7. It is a homomorphism. Let $a, b \in G_i$. Then

$$\begin{aligned} \phi(ab) &= (e_1, e_2, \cdots, ab, \cdots, e_r) \\ &= (e_1, e_2, \cdots, a, \cdots, e_r)(e_1, e_2, \cdots, b, \cdots, e_r) \\ &= \phi(a)\phi(b). \end{aligned}$$

8. It is not a homomorphism if G is not abelian. We have $\phi(ab) = (ab)^{-1} = b^{-1}a^{-1} = \phi(b)\phi(a)$ which may not equal $\phi(a)\phi(b)$ if G is not abelian. For a specific example, let $G = S_3$ with our the notation in Section 8. Then $\phi(\rho_1\mu_1) = \phi(\mu_3) = {\mu_3}^{-1} = \mu_3$, but $\phi(\rho_1)\phi(\mu_1) = {\rho_1}^{-1}{\mu_1}^{-1} = \rho_2\mu_1 = \mu_2$.

9. Yes, it is a homomorphism. By calculus, $(f+g)'' = f'' + g''$. Then $\phi(f+g) = (f+g)'' = f'' + g'' = \phi(f) + \phi(g)$.

10. Yes, it is a homomorphism since we have $\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$, so $\phi(f+g) = \int_0^4 [f(x)+g(x)]\,dx = \int_0^4 f(x)\,dx + \int_0^4 g(x)\,dx = \phi(f) + \phi(g)$.

11. Yes, it is a homomorphism. By definition, $3(f+g)(x) = 3[f(x)+g(x)] = 3\cdot f(x) + 3\cdot g(x) = (3f)(x) + (3g)(x) = (3f+3g)(x)$, showing that $3(f+g)$ and $3f+3g$ are the same function. Thus $\phi(f+g) = 3(f+g) = 3f+3g = \phi(f) + \phi(g)$.

12. No, it is not a homomorphism. Let $n = 2$ and $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, so that $A + B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. We see that $\phi(A+B) = \det(A+B) = 4 - 1 = 3$ but $\phi(A) + \phi(B) = \det(A) + \det(B) = 1 + 0 = 1$.

13. Yes, it is a homomorphism. Let $A = (a_{ij})$ and $B = (b_{ij})$ where the element with subscript ij is in the ith row and jth column. Then

$$\begin{aligned} \phi(A+B) &= \operatorname{tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} \\ &= \operatorname{tr}(A) + \operatorname{tr}(B) = \phi(A) + \phi(B). \end{aligned}$$

14. No, it is not a homomorphism. We see that $\phi(I_n I_n) = \phi(I_n) = \operatorname{tr}(I_n) = n$, but $\phi(I_n) + \phi(I_n) = \operatorname{tr}(I_n) + \operatorname{tr}(I_n) = n + n = 2n$.

15. No, it is not a homomorphism. Let $f(x) = x^2 + 1$. We have $\phi(f\cdot f) = \int_0^1 (x^2+1)^2\,dx = \int_0^1 (x^4 + 2x^2 + 1)\,dx = \frac{1}{5} + \frac{2}{3} + 1 = \frac{28}{15}$ but $\phi(f)\phi(f) = \left[\int_0^1 (x^2+1)\,dx\right]^2 = (\frac{1}{3}+1)^2 = \frac{16}{9}$.

16. $\operatorname{Ker}(\phi)$ consists of the even permutations, so $\operatorname{Ker}(\phi) = A_3 = \{\rho_0, \rho_1, \rho_2\}$.

17. $\operatorname{Ker}(\phi) = 7\mathbb{Z}$ because 4 has order 7 in $\mathbb{Z}_7$. We have

$$\begin{aligned} \phi(25) &= \phi(21+4) = \phi(21) +_7 \phi(4) = 0 +_7 \phi(4) \\ &= \phi(1) +_7 \phi(1) +_7 \phi(1) +_7 \phi(1) \\ &= 4 +_7 4 +_7 4 +_7 4 = 1 +_7 1 = 2. \end{aligned}$$

18. $\operatorname{Ker}(\phi) = 5\mathbb{Z}$ because 6 has order 5 in $\mathbb{Z}_{10}$. We have

$$\begin{aligned} \phi(18) &= \phi(15+3) = \phi(15) +_{10} \phi(3) = 0 +_{10} \phi(3) \\ &= \phi(1) +_{10} \phi(1) +_{10} \phi(1) = 6 +_{10} 6 +_{10} 6 \\ &= 2 +_{10} 6 = 8. \end{aligned}$$

19. In S_8, we have $\sigma = (1,4,2,6)(2,5,7) = (1,4,2,5,7,6)$ which is of order 6, so $\operatorname{Ker}(\phi) = 6\mathbb{Z}$. Then $\phi(20) = \phi(18+2) = \phi(18)\phi(2) = \iota\,\sigma^2 = (1,2,7)(4,5,6)$.

20. $\operatorname{Ker}(\phi) = \langle 5\rangle = \{0,5\}$ because 8 has order 5 in $\mathbb{Z}_{20}$. We have $\phi(3) = 8 +_{20} 8 +_{20} 8 = 16 +_{20} 8 = 4$.

21. The element $\sigma = (2,5)(1,4,6,7)$ has order 4, so $\operatorname{Ker}(\phi) = \langle 4\rangle = \{0,4,8,12,16,20\}$. Then $\phi(14) = \phi(12 +_{24} 2) = \iota\,\sigma^2 = (1,6)(4,7)$.

22. Now $\phi(m,n) = 3m-5n$ so $\text{Ker}(\phi) = \{(m,n) \mid 3m = 5n \text{ for } m,n \in \mathbb{Z}\}$. Then $\phi(-3,2) = 3(-3)-5(2) = -19$.

23. We have $\phi(m,n) = (2m-n, -3m+5n)$ and the only simultaneous solution of the equations $2m-n=0$ and $-3m+5n=0$ is $m=n=0$, so $\text{Ker}(\phi) = \{(0,0)\}$. Also, $\phi(4,6) = (8-6, -12+30) = (2,18)$.

24. Let $\sigma = (3,5)(2,4)$ and $\mu = (1,7)(6,10,8,9)$. Because σ has order 2 and μ has order 4, we see that $\text{Ker}(\phi) = 2\mathbb{Z} \times 4\mathbb{Z}$. Because our all the cycles are disjoint, we find that

$$\begin{aligned}\phi(3,10) &= \sigma^3\mu^{10} = (3,5)^3(2,4)^3(1,7)^{10}(6,10,8,9)^{10} \\ &= (3,5)(2,4)(6,10,8,9)^2 = (3,5)(2,4)(6,8)(9,10).\end{aligned}$$

25. Because the homomorphism ϕ must be *onto* $\mathbb{Z}$, $\phi(1)$ must be a generator of $\mathbb{Z}$. Thus there are only two such homomorphisms ϕ, one where $\phi(1) = 1$ so $\phi(n) = n$ for all $n \in \mathbb{Z}$, and one where $\phi(1) = -1$ so $\phi(n) = -n$ for all $n \in \mathbb{Z}$.

26. There are an infinite number of them. For any nonzero $n \in \mathbb{Z}$, we know that $\langle n \rangle$ is isomorphic to $\mathbb{Z}$, and that $\phi : \mathbb{Z} \to \mathbb{Z}$ given by $\phi(m) = mn$ is an isomorphism, and hence a homomorphism. Of course ϕ defined by $\phi(m) = 0$ for all $m \in \mathbb{Z}$ is also a homomorphism.

27. There are two of them; one where $\phi(1) = 1$ (see Example 13.10 with n = 2) and one where $\phi(1) = 0$.

28. Because we must have $\phi_g(e)$ by Theorem 13.12, we must have $ge = e$, so $g = e$ is the only possibility. Because $\phi_e(x) = ex = x$ is the identity map, it is indeed a homomorphism.

29. We have $\phi_g(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \phi_g(x)\phi_g(y)$ for all $x, y \in G$, so ϕ_g is a homomorphism for all $g \in G$.

30. Incorrect. It should say what ϕ maps to what, what x and y are, and include the necessary quantifier, "for all".

A map ϕ of a group G into a group G' is a **homomorphism** if and only if $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$.

31. The definition is correct. **32.** T T F T F F T T F F

33. There are no nontrivial homomorphisms. By Theorem 13.12, the image $\phi[\mathbb{Z}_{12}]$ would be a subgroup of $\mathbb{Z}_5$, and hence all of $\mathbb{Z}_5$ for a nontrivial ϕ. By Theorem 13.15, the number of cosets of $\text{Ker}(\phi)$ must then be 5. But the number of cosets of a subgroup of a finite group is a divisor of the order of the group, and 5 does not divide 12.

34. Let $\phi(n)$ be the remainder of n when divided by 4 for $n \in \mathbb{Z}_{12}$. Replacing 6 by 12 and 2 by 4 in the solution of Exercise 4 shows that ϕ is a homomorphism.

35. Let $\phi(m,n) = (m,0)$ for all $(m,n) \in \mathbb{Z}_2 \times \mathbb{Z}_4$.

36. There are no nontrivial homomorphisms because $\mathbb{Z}$ has no finite subgroups other than $\{0\}$.

37. Let $\phi(n) = \rho_n$ for $n \in \mathbb{Z}_3$, using our notation in Section 8 for elements of S_3. Both $\mathbb{Z}_3$ and $\langle \rho_1 \rangle$ are cyclic of order 3.

38. Let $\phi(n)$ be the identity in S_3 for n even, and the transposition (1,2) for n odd in $\mathbb{Z}$. Note that $\langle (1,2) \rangle$ is of order 2, isomorphic to $\mathbb{Z}_2$, and this homomorphism mirrors the homomorphism γ of Example 13.10 for $n = 2$.

39. Let $\phi(m,n) = 2m$. Then $\phi((m,n)+(r,s)) = \phi(m+r, n+s) = 2(m+r) = 2m+2r = \phi(m,n)+\phi(r,s)$.

40. Let $\phi(2n) = (2n, 0)$ for $n \in \mathbb{Z}$. Then $\phi(2m+2n) = \phi(2(m+n)) = (2(m+n), 0) = (2m+2n, 0) = (2m, 0) + (2n, 0) = \phi(2n) + \phi(2m)$.

41. Viewing D_4 as a group of permutations, let $\phi(\sigma) = (1,2)$ for each odd permutation $\sigma \in D_4$, and let $\phi(\sigma)$ be the identity permutation for each even $\sigma \in D_4$. Note that $\langle (1,2) \rangle$ is a subgroup of S_3 of order 2, isomorphic to $\mathbb{Z}_2$. This homomorphism mirrors the homomorphism for $n = 4$ in Example 13.3, restricted to the subgroup D_4 of S_4.

42. For each $\sigma \in S_3$, let $\phi(\sigma) = \mu$ where $\mu(i) = \sigma(i)$ for $i = 1, 2, 3$ and $\mu(4) = 4$. This is obviously a homomorphism.

43. Let $\phi(\sigma) = (1,2)$ for each odd permutation $\sigma \in S_4$, and let $\phi(\sigma)$ be the identity permutation for each even $\sigma \in S_4$. Note that $\langle (1,2) \rangle$ is a subgroup of S_3 of order 2, isomorphic to $\mathbb{Z}_2$. This homomorphism mirrors the homomorphism for $n = 4$ in Example 13.3.

44. Because $\phi[G] = \{\phi(g) \mid g \in G\}$, we see that $|\phi[G]| \leq |G|$, so $|\phi[G]|$ must be finite also. By Theorem 13.15, there is a one-to-one correspondence between the elements of $\phi[G]$ and the cosets of $\text{Ker}(\phi)$ in G. Thus $|\phi[G]| = |G|/|\text{Ker}(\phi)|$, so $|\phi[G]|$ divides $|G|$.

45. By Theorem 13.14, $\phi[G]$ is a subgroup of G', so if $|G'|$ is finite, then $|\phi[G]|$ is finite. By the Theorem of Lagrange, we see that $|\phi[G]|$ is then a divisor of $|G'|$.

46. Let $x \in G$. By Theorem 7.6, there are (not necessarily distinct) indices $i_1, i_2, i_3, \cdots, i_m$ in I such that

$$x = a_{i_1}^{n_1} a_{i_2}^{n_2} a_{i_3}^{n_3} \cdots a_{i_m}^{n_m} \text{ where the } n_j \text{ are in } \mathbb{Z}.$$

Because $\phi(a_{i_j}) = \mu(a_{i_j})$ for $j = 1, 2, 3, \cdots, m$, it follows from Definition 13.1 (extended by induction) and Property 2 in Theorem 13.12 that $\phi(x) = \mu(x)$. Thus ϕ and μ are the same map of G into G'.

47. By Theorem 13.12, $\text{Ker}(\phi)$ is a subgroup of G. By the Theorem of Lagrange, either $\text{Ker}(\phi) = \{e\}$ or $\text{Ker}(\phi) = G$ because $|G|$ is a prime number. If $\text{Ker}(\phi) = \{e\}$, then ϕ is one to one by Corollary 13.18. If $\text{Ker}(\phi) = G$, then ϕ is the trivial homomorphism, mapping everything into the identity element.

48. We see that $\text{Ker}(\text{sgn}_n) = A_n$. The multiplicative group $\{-1, 1\}$ is isomorphic to the group $\mathbb{Z}_2$, and if $1 \in \{-1, 1\}$ is renamed 0 and -1 is renamed 1, then this becomes the homomorphism of Example 13.3.

49. Let $a, b \in G$. For the composite function $\gamma\phi$, we have

$$\gamma\phi(ab) = \gamma(\phi(ab)) = \gamma(\phi(a)\phi(b)) = \gamma(\phi(a))\gamma(\phi(b)) = \gamma\phi(a)\gamma\phi(b)$$

where the first equality uses the definition of the composite map $\gamma\phi$, the second equality uses the fact that ϕ is a homomorphism, the third uses the fact that γ is a homomorphism, and the last uses the definition of $\gamma\phi$ again. This shows that $\gamma\phi$ is indeed a homomorphism.

50. Let $x', y' \in \phi[G]$ and let $\phi(x) = x'$ and $\phi(y) = y'$ where $x, y \in G$. Then $\phi[G]$ is abelian

$$\begin{array}{l}
\text{if and only if } x'y' = y'x', \\
\text{if and only if } (y'x')^{-1}x'y' = e', \\
\text{if and only if } x'^{-1}y'^{-1}x'y' = e', \\
\text{if and only if } \phi(x)^{-1}\phi(y)^{-1}\phi(x)\phi(y) = e', \\
\text{if and only if } \phi(x^{-1}y^{-1}xy) = e', \\
\text{if and only if } x^{-1}y^{-1}xy \in \text{Ker}(\phi)
\end{array}$$

for all $x', y' \in \phi[G]$. Note that because x' and y' could be any elements of $\phi[G]$, x and y could be any elements of G.

51. Let $m, n \in \mathbb{Z}$. We have $\phi(m+n) = a^{m+n} = a^m a^n = \phi(m)\phi(n)$, showing that ϕ is a homomorphism. The image of ϕ is the cyclic subgroup $\langle a \rangle$ of G, and $\text{Ker}(\phi)$ is one of the subgroups of $\mathbb{Z}$, which must be cyclic and consist of all (positive, negative and zero) multiples of some integer j in $\mathbb{Z}$. If a has finite order in G, then j is the order of a; otherwise, $j = 0$.

52. We show that each of $S = \{x \in G \mid \phi(x) = \phi(a)\}$ and Ha is a subset of the other. Let $s \in S$. Using Theorem 13.12 and the homomorphism property, we have $\phi(sa^{-1}) = \phi(s)\phi(a^{-1}) = \phi(a)\phi(a^{-1}) = \phi(a)\phi(a)^{-1} = e'$ so $sa^{-1} = h \in H$. Then $s = ha$ so $s \in Ha$. Thus S is a subset of Ha.

Let $h \in H$ so that $ha \in Ha$. Then $\phi(ha) = \phi(h)\phi(a) = e'\phi(a) = \phi(a)$, showing that $ha \in S$. Thus Ha is a subset of S, so $Ha = S$.

53. We have $\phi(1,0) = h^1k^0 = h$ and $\phi(0,1) = h^0k^1 = k$. Let ϕ be a homomorphism. Using addition notation in $\mathbb{Z} \times \mathbb{Z}$ as usual, we have

$$\begin{aligned} \phi(1,1) &= \phi((1,0)+(0,1)) = \phi(1,0) + \phi(0,1) = hk, \\ \phi(1,1) &= \phi((0,1)+(1,0)) = \phi(0,1) + \phi(1,0) = kh. \end{aligned}$$

Thus if ϕ is a homomorphism, we must have $hk = kh$.

Conversely, suppose that $hk = kh$. Then for any (i,j) and (m,n) in $\mathbb{Z} \times \mathbb{Z}$, we have

$$\begin{aligned} \phi((i,j)+(m,n)) &= \phi(i+m, j+n) = h^{i+m}k^{j+n} = h^i h^m k^j k^n \\ &= h^i k^j h^m k^n = \phi(i,j)\phi(m,n) \end{aligned}$$

where the first equality in the second line follows from the commutativity of h and k. Thus ϕ is a homomorphism if and only if $hk = kh$.

54. The preceding exercise shows that ϕ is a homomorphism for all choices of h and k in G if and only if $hk = kh$ for all h and k in G, that is, if and only if G is an abelian group.

55. The map ϕ is a homomorphism if and only if $h^n = e$, the identity in G.

Proof: If ϕ is a homomorphism, then $\phi(0) = e$. Consequently

$$h^n = \phi(1)^n = \phi(\underbrace{1+1+\cdots+1}_{n \text{ summands}}) = \phi(0) = e.$$

Conversely, suppose that $h^n = e$, so that $\langle h \rangle \simeq \mathbb{Z}_m$ where m is a divisor of n. Let $i, j \in \mathbb{Z}_n$. Viewing i and j in $\mathbb{Z}$, write $i+j = qm+r$, $i = q_1m+r_1$, and $j = q_2m+r_2$, all by the division algorithm. Then $\phi(i+j) = h^{i+j} = h^{qm+r} = (h^m)^q h^r = e^q h^r = h^r$. Similarly, $\phi(i) = h^{r_1}$ and $\phi(j) = h^{r_2}$, so $\phi(i)\phi(j) = h^{r_1+r_2}$. Because $i+j = (q_1+q_2)m + r_1 + r_2$, the remainder $i+j$ when divided by m is the same as the remainder of r_1+r_2 when divided by m. Thus $h^{r_1+r_2} = h^r$ so $\phi(i)\phi(j) = \phi(i+j)$. Hence ϕ is a homomorphism.